
QML for detecting AMR from EHR

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ABSTRACT

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Keywords First keyword · Second keyword · etc.

1 Introduction

We can detect antimicrobial resistance in patients given electronic health records (EHRs). [1] This is useful because EHRs are available for the vast majority of patients for little to no additional cost. This dataset has been used to predict antimicrobial resistance using classical machine learning methods, [2] but not yet for shapley value analysis or quantum machine learning. Other research has demonstrated that this can be done using qml from urine samples, [3] but even that cost may be too high or inaccessible in some areas.

We want to contribute to the literature around the efficacy of quantum machine learning to learn what domains it is most useful in. [4] We do this by comparing performance to a baseline classical model, with separate models for each antibiotic. This allows us to create a fine-grained picture of how the quantum model performs in relation to the classical model.

We also want to use a practical application of shapley values for qml. This has been done for other applications in medicine [5] but we want to see how each feature impacts the result for EHRs. This has serious implications for social factors of disease prevention, as it would be possible to identify which specific aspects of a patient's health record contribute most to an identification of antimicrobial-resistant bacteria and allow public

health policy makers to pre-emptively support people in communities or areas that may be most at risk. [6]

2 Methods

Several features were extracted from the raw dataset. They are Area Deprivation Index (ADI), Age, Gender, time (days) since last nursing home visit, time (days) since last dose of antibiotic, most recent procedure, time (days) since most recent procedure, the type of ward the patient was seen in, and a list of all comorbidities each patient had. This totalled 520 features. These features were chosen to train models that could be effective given minimally expensive information, not requiring microbiology testing or body sample collection. The dataset was then segregated by antibiotic, and only those antibiotics with subjects between 4500 and 11000 subjects were used (for a total of 9 antibiotics), balancing generalizability and size so training could be completed in a reasonable amount of time.

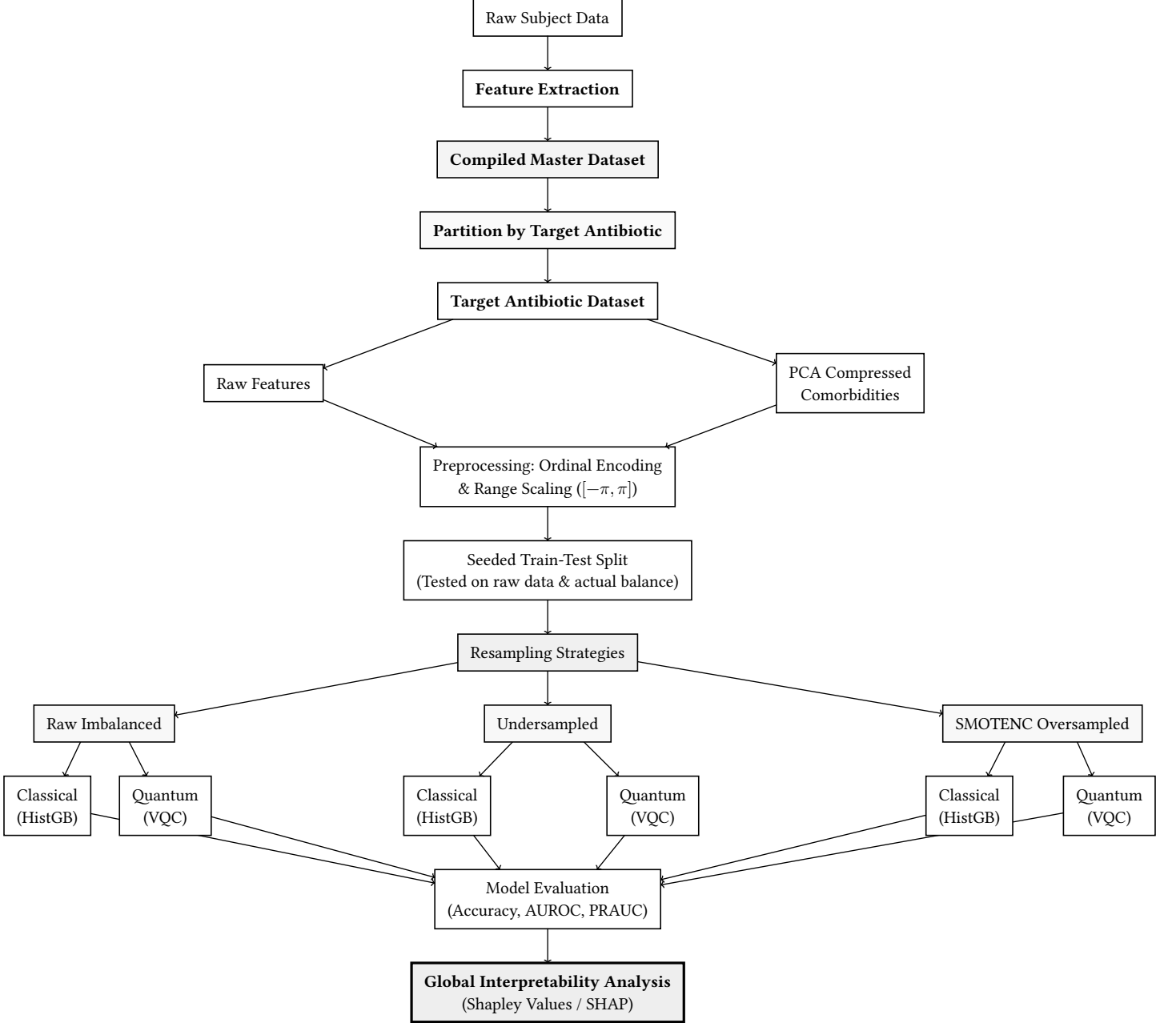


Figure 1: End-to-end data processing, resampling, and dual-architecture modeling pipeline.

We used a Histogram-based Gradient Boosting Classifier from the scikit-learn package to serve as our classical baseline. It is based on the Gradient Boosting Classifier, which was chosen because of its high accuracy and performance on similar datasets in the medical field. [7] The Histogram-based version of the model was chosen for its performance on large datasets. [8]

While the classical model could run on the full set of 520 features, the quantum model could not. So, PCA from the scikit-learn package was used to reduce the number of features being used by the quantum circuit. The vast majority of the 520 features (508) were the comorbidities of each patient, so these were reduced by PCA to 12 features, for a total of 20 features. A standard list of comorbidity groupings was not used, in order to identify latent correlations within our specific dataset without any potential bias from existing groupings.

Because the data was severely imbalanced on our target with varying imbalance for each antibiotic, we also implemented undersampling and SMOTENC oversampling using the

imbalanced-learn package to balance the dataset, training different models for each situation. In total, for the classical models we had 6 models for each antibiotic (raw, oversampled, and undersampled for 521 and 20 features) and for the quantum models we had 3 (raw, oversampled, and undersampled).

We used a Variational Quantum Circuit (VQC) implemented using the qiskit-machine-learning package. We used the Aer Sampler in statevector mode with GPU acceleration for the sampler. We used the EfficientSU2 circuit from qiskit for the feature map with 1 repetition, such that 4 features were mapped to a qubit. The EfficientSU2 circuit was constructed with full entanglement. We used the RealAmplitudes circuit from qiskit for the ansatz with 3 repetitions to maximize expressibility without increasing circuit depth too much. We used COBYLA from qiskit-machine-learning for the optimizer with 100 maximum iterations.

From each model we gathered its accuracy score, a confusion matrix, Area Under the Receiver Operating Characteristic Curve (AUROC), and Precision-Recall Area Under the Curve (PRAUC). SHAP values were also calculated for each model.

3 Results

4 Discussion

5 Conclusion

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